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Operational culture conditions determinate benzalkonium chloride resistance in *L. monocytogenes-E. coli* dual species biofilms

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A R T I C L E I N F O

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A B S T R A C T

Biofilms pose a serious challenge to the food industry. Higher resistance of biofilms to any external stimuli is a major hindrance for their eradication. In this study, we compared the growth dynamics and benzalkonium chloride (BAC) resistance of dual species *Listeria monocytogenes-Escherichia coli* 48 h biofilms formed on stainless steel (SS) coupons surfaces under *batch* and *fed-batch* cultures. Differences between both operational culture conditions were evaluated in terms of total viable adhered cells (TVAC) in the coupons during 48 h of the mixed-culture and of reduction of viable adhered cells (RVAC) obtained after BAC-treatment of a 48 h biofilm of L. *monocytogenes-E. coli* formed under both culture conditions. Additionally, epifluorescence microscopy (EFM) and confocal scanning microscopy (CLSM) permitted to visualize the 2D and 3D biofilms structure, respectively. Observed results showed an increase in the TVAC of both strains during biofilm development, being the number of *E. coli* adhered cells higher than L. *monocytogenes* in both experimental systems ($p < 0.05$). Additionally, the number of both strains were higher approximately 2.0 log CFU/coupon in *batch* conditions compared to *fed-batch* system ($p < 0.05$). On the contrary, significantly higher resistance to BAC was observed in biofilms formed under *fed-batch* conditions. Furthermore, in *batch* system both strains had a similar reduction level of approximately 2.0 log CFU/coupon, while significantly higher resistance of *E. coli* compared to L. *monocytogenes* (reduction level of 0.69 and 1.72 log CFU/coupon, respectively) ($p < 0.05$) was observed in *fed-batch* system. Microscopic image visualization corroborated these results and showed higher complexity of 2D and 3D structures in dual species biofilms formed in *batch* cultures. Overall, we can conclude that the complexity of the biofilm structure does not always imply higher resistance to external *stimuli*, and highlights the need to mimic industrial operational conditions in the experimental systems in order to better assess the risk associated to the presence of pathogenic bacterial biofilms.

1. Introduction

Biofilms are a serious challenge for the food industry. Their appearance in food-processing environment creates a risk of food contamination, thus causing a threat to the entire food chain (Srey et al., 2013). The most undesirable are biofilms formed by bacterial pathogens in food industry surfaces, which can reach humans though consumption of contaminated food after cross-contamination (Bridier et al., 2015; Simões and Simões, 2010). Food contaminated with pathogens such as *Listeria monocytogenes* can potentially be fatal to YOPI (young, old, pregnant, immunocompromised) consumers (Jeyaletchumi et al., 2010).

The elimination of bacterial pathogens from the food-processing environment is particularly difficult due to their ability to form biofilms. Many previous studies showed that biofilms are more resistant to different antimicrobial substances, including disinfectants used to cleaning and disinfection in food industry (Amalaradjou et al., 2009; Chaitiemwong et al., 2010; Cruz and Fletcher, 2011; Rodriguez-Lopez and López-Cabo, 2017). Food-processing environments have a favorable conditions for biofilms development, because there are nutrients and various surface locations such as gouges, scratches, pits, cracks that facilitates cells adhesion and growth (Iñiguez-Moreno et al., 2019).

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In the context of food industry, *L. monocytogenes* is one of the microorganisms of food safety concern, causing listeriosis with high mortality rate (EFSA, 2015; Rodriguez-Lopez et al., 2015). Indeed, *L. monocytgenes* is considered a weak biofilm former, but in co-existence with other species, the ability to biofilm formation increased (Mosquera-Fernández et al., 2014). Interspecies interactions occurring between cohabiting strains in the same biofilm are not the only factor affecting the number of cells and their resistance, the conditions in which the biofilm is formed are also important, such as access to nutrients (Van der Veen and Abee, 2011). Another problem with the elimination of L. *monocytogenes* from food-processing plants is that it can persist for months or even years on equipment items, in drains or on the floor (Ferreira et al., 2014). In addition, *L. monocytogenes* is characterized by the ability to survive in an environment of food processing, i.e. refrigerated temperatures, heat, desiccation or the presence of a high amount of salt, whereas for other microorganisms these conditions are limiting factors (Gardan et al., 2003; Zoz et al., 2017).

The study of biofilms is even more complex if we take into account that biofilms can be found in very different locations along the food-processing plants (Rodriguez-Lopez and López-Cabo, 2017). Biofilms are formed in spaces of different geometry, different external conditions and exposed or not to different hydrodynamic conditions caused by continuous and discontinuous flow of liquids such as water, milk or juices, depending on the food sector. However, the number of articles comparing biofilms formed under different modalities of culture that could mimic different industrial conditions is limited (Cunault et al., 2015; Le Gentil et al., 2010). Most of the available research focuses on comparing mono- and dual- or multi-species biofilms formed in different surfaces present in food processing plants under *batch* conditions (Kocot and Olszewska, 2019; Rodriguez-Lopez and López-Cabo, 2017; Van der Veen and Abee, 2011). The importance of different operational conditions and the inclusion of these parameters in the research results e.g. from the fact that L. *monocytognes*-carrying contamination sites were found in very different locations of a food-processing plant (Rodríguez-López et al., 2020).

Therefore, in this study, we compared growth dynamics, BAC resistance and microscopic 2D and 3D structures of L. *monocytogenes-E. coli* biofilms formed on SS coupons under *batch* and *fed-batch* culture conditions.

2. Materials and methods

2.1. Bacterial strains

L. monocytogenes A1 and *E. coli* A14 used in this study were previously isolated from a fish industry processing plant and genetically characterized (Rodriguez-Lopez and López-Cabo, 2017). Stock cultures were maintained at −80 °C in brain heart infusion broth (BHI; Biolife, Milan, Italy) with 50% glycerol (1:1, *v*/v). Working cultures were prepared similarly but maintained at −20 °C until use.

2.2. Inocula preparation

L. monocytogenes A1 and *E. coli* A14 stock working cultures were activated by twice cultured (100 μL each time) in 5 mL of TSB (Tryptic Soy Broth, Cultimed, S.L., Spain) at 37 °C. In both species, inocula was prepared by adjusted activated cultures (Abs_{700}) to 0.1 ± 0.001 with using sterile PBS and spectrophotometer (Cecil Instruments, Cambridge, England) to obtain a bacterial concentration of~8 log CFU/mL accordingly with previous calibrations. Adjusted cultures were further diluted in TSB to obtain a final concentration of each species of~ 4log CFU/mL and used as inoculum for dual-species biofilm formation on SS coupons surface under *batch* and *fed-batch* conditions.

2.3. Dual-species biofilms formation

SS coupons (Ø 12.7 mm) (Comevisa, Vigo, Spain) were washed with a detergent solution for 10 min and next with 70% ethanol (*v*/v) for 1 min in ultrasonic cleaner (BRANSOR 250, Thermo Fisher, USA), rinsed with deionized water, air-dried and autoclaved at 121 °C for 20 min. Cleaned coupons were disposed in both experimental systems as follows:

i) *Batch* system: sterilized coupons were individually placed in 24-well polystyrene flat-bottomed plates (Falcon, Corning, NY, USA) and inoculated with 1 mL of adjusted co-culture of L. *monocytogenes* and *E. coli* and incubated at 25 °C for 48 h.
ii) *Fed-batch* system; CDC bioreactor (Fig. 1): sterilized coupons were placed in coupons holder available in the CDC biofilm reactor. Initially, 3 mL of inocula were added to 300 mL of TSB and culture was maintained under static conditions. After 3 h the CDC culture was performed under stirring (60 rpm) until the end of the experiment. *Fed batch* cultivation was performed by continuously fluxing of 3 mL of TSB to maintain glucose concentration accordingly with preliminary experiments (data not shown). The CDC culture was incubated at 25 °C for 48 h.

Coupons were removed from both experimental systems for TVAC and RTVC determinations. For the experiments carried out to determine the kinetic of growth of both species (Section 2.4.), coupons were removed after 8, 12,17, 24, 36 and 48 h of culturing. BAC-resistance experiments (Section 2.5) were carried out with coupons removed after 48 h culturing.

2.4. Determination of total viable adhered cells (TVAC) in the SS coupons

First, each coupon was gently rinsed with PBS to eliminate non-adhered cells and then placed in Falcon tube with 1 mL of buffered peptone water (BPW; Cultimed, Barcelona, Spain). Tubes were vortexed for 1 min and then sonicated for another 1 min (38 kHz, 30 °C) (BRANSOR 250, Thermo Fisher, USA). Finally, samples were serially diluted in BPW and in duplicate spread-plated on ALOA (Cultimed, Barcelona, Spain) and HiCrome™ Coliform agar (Sigma-Aldrich, St Louis, MO, USA) to quantify L. *monocytogenes* and *E. coli*, respectively. Plates were incubated at 37 °C for 48 h. Automatic counting was performed by using Flash & Go Automatic Colony Counter (SCAN 500, Interscience). TVAC were expressed in log CFU/coupon (cm^2). Experiments were done in triplicate.

2.5. Resistance to BAC of L. monocytogenes-E. coli biofilms formed under batch and fed-batch conditions

2.5.1. BAC and neutralizing solution preparation

Benzalkonium chloride (BAC, Guinama, Alboraya, Spain) was prepared at 200 ppm by dissolving the stock solution in sterile distilled water and kept at 4 °C until use. Neutralizing solution was prepared with following composition, per liter: 10 ml of a 34 g l^{-1} KH_2PO_4 solution adjusted to pH = 7.2 with NaOH ($_{aq}$), 3 g soy lecithin, 5 g $Na_2S_2O_3$, 1 g L-histidine, 30 mL Tween 80 and deionized water (Rodriguez-Lopez et al., 2017). This solution was sterilized by autoclaving at 121 °C for 20 min and kept at 4 °C until use.

2.5.2. BAC treatment

The effect of 200 ppm of BAC against 48 h old L. *monocytogenes-E. coli* biofilms formed under both experimental conditions was assayed. SS coupons removed from both culture conditions were placed separately in another 24-bottom well microtiter plates. Next, 1 mL of 200 ppm BAC solution was added to each well and allowed to remain for 10 min at room temperature. Treated coupons were transferred to new wells containing 1 mL of neutralizing solution and immersed for 30 s, which was considered the minimum time necessary for proper neutralization

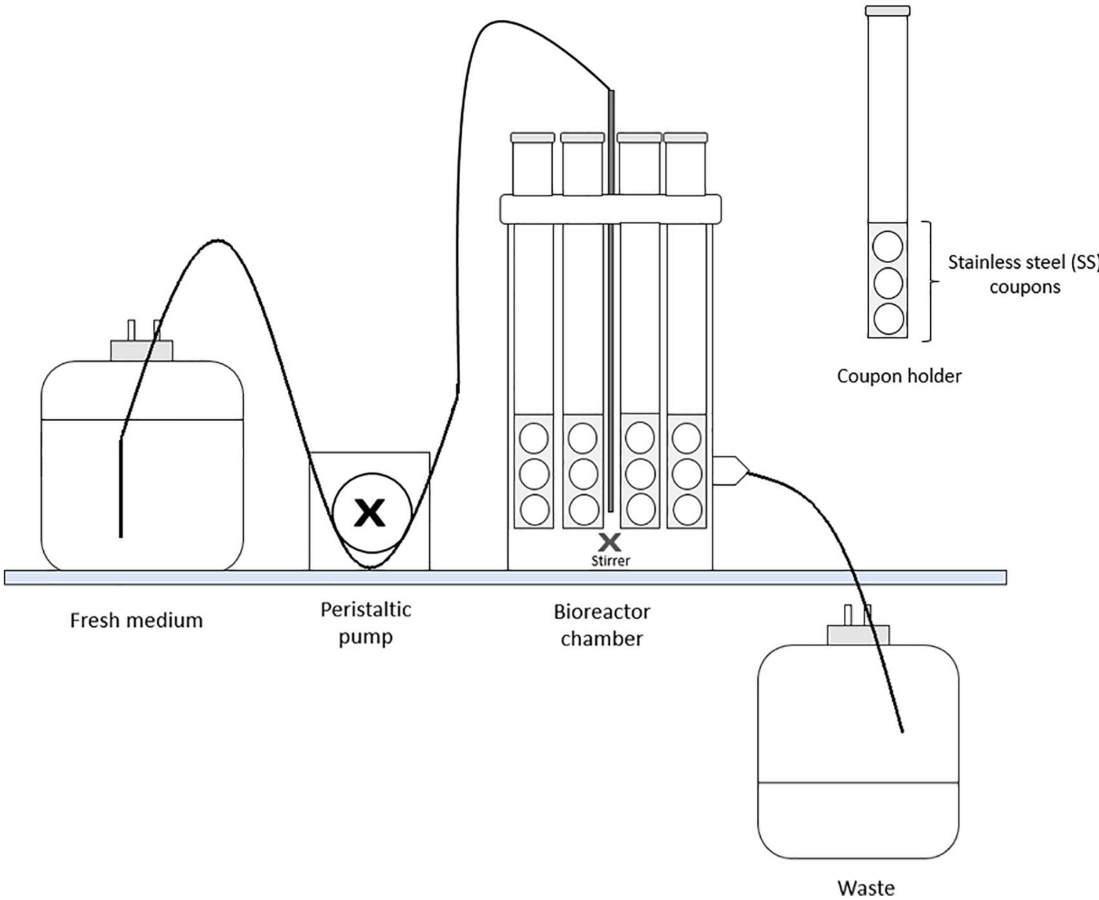

Fig. 1. The scheme showing the CDC bioreactor presenting the setup of the *fed-batch* experiment.

according to a previous assay. Untreated biofilm samples were used as controls. Finally, TVAC in controls and treated samples was calculated as described above and RVAC was calculated as the difference between TVAC present in mixed biofilms formed in SS coupons in controls and after BAC treatment. RVAC is expressed in log CFU/coupon.

2.6. Microscopic assays

The 48 h *L. monocytogenes-E. coli* biofilms from both experimental systems before and after BAC-inactivation were rinsed with 1 mL of 0,9% NaCl and stained for examination under epifluorescence (EFM) and confocal scanning microscopy (CLSM).

For EFM, biofilms were stained with ViaGram ™ Red^{+} Bacterial Gram Stain and Viability Kit (Life Technologies, Eugene, OR) according to the manufacturer's instructions. This stain allows in a mixed population the distinction of Gram-positive cells (red emitting) with a counterstain with DAPI. After incubation, the staining mixture was removed and coupons were mounted on microscopic slides with using one drop of mounting medium (FluoroShield; Life Technologies, Eugene, OR). Microscopic visualization was conducted with a Leica DM 6000 epifluorescence microscope (Leica, Wetzlar, Germany) using a 40× objective and 10× ocular lens Metamorph MMAF software (Molecular Devices, Sunnyvale, CA, USA).

For CLSM, biofilms were stained with LIVE/DEAD® viability kit (Life Technologies, Eugene, OR, USA) following manufacturer instructions in order to observe the final biofilm morphology and the distribution of live (green-emitting) cells and damaged/dead (red-emitting) cells. Next, stained biofilms were fixed with 1 mL of 4.0% formaldehyde (Sigma-Aldrich, Lisbon, Portugal) and incubated for 30 min at room temperature. Then, the samples were washed with PBS and then treated with 1 mL of 50 mM ammonium chloride and incubated for 3 min at room temperature. Fixed biofilms were placed on microscopic slides covered with coverslips using Baclight immersion oil (ThermoFisher Scientific, Massachusetts, USA). Confocal images were acquired using an upright laser scanning microscope SPE with a Plan-apochromatic 63×/NA 1.4 objective (Leica Microsystems, Wetzlar, Germany). Green fluorescence from Syto 9 was acquired exciting samples with 488 nm laser and emission signal was detected between 490 and 550 nm. Red fluorescence from PI was acquired exciting samples with 561 nm laser and emission signal was detected between 590 and 650 nm. All images shown correspond to a maximal intensity projection of a confocal z-stack. Images were processed with Adobe Photoshop CS.

2.7. Statistical analysis

Statistical analysis was conducted with the use Statistica software ver. 13.1 (StatSoft Inc., Tulsa, OK). Inactivated samples were tested by one-way ANOVA test. Differences were considered significant at $p <$ 0.05 level of probability.

3. Results

3.1. Kinetics of growth of L. monocytogenes-E. coli under batch and fed-batch culture

Dynamics of biofilm formation on SS coupons in *batch* and *fed-batch* conditions are shown in Fig. 2. The number of cells of both strains increased in the following hours of biofilm growth, regardless of the experimental conditions and after 48 h of biofilm development reached the values of 6.98 and 8.03 log CFU/coupon and 4.98 and 6.35 log CFU/

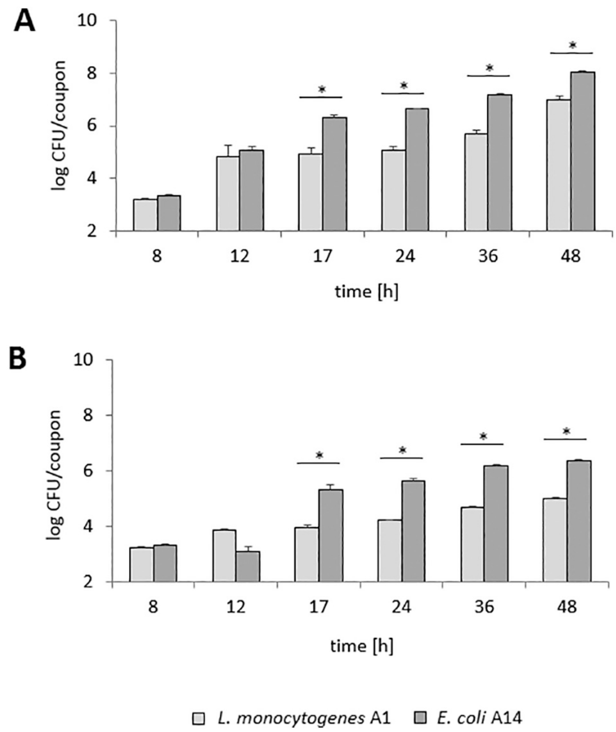

Fig. 2. The cell counts of *L. monocytogenes* A1 (▫) and *E. coli* A14 (▪) in dual-species biofilms formed in *batch* (A) and *fed-batch* (B) system for 48 h. The bars represent the mean values ± standard deviations. Asterisks indicate significant differences between counts of L. *monocytogenes* A1 and *E. coli* A14 ($p < 0.05$).

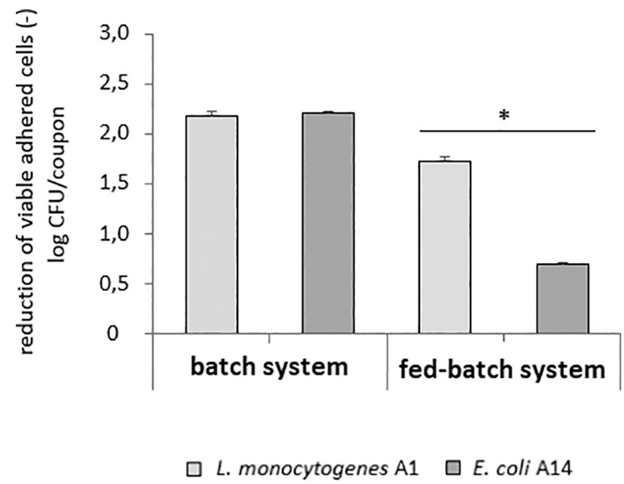

Fig. 3. The effect of BAC-based disinfectant on inactivation of 48 h old dual-species biofilm of *L. monocytogenes* A1 (▫) and *E. coli* A14 (▪) formed under *batch* and *fed-batch* conditions after 10 min of BAC treatment. The bars represent the mean values ± standard deviations. Asterisks indicate significant differences between log reduction levels of L. *monocytogenes* A1 and *E. coli* A14 ($p < 0.05$).

coupon for L. *monocytogenes* and *E. coli* under *batch* and *fed-batch* conditions, respectively. It was shown that the number of cells of both strains had higher numbers under *batch* conditions as compared to the *fed-batch* system – these differences were about 2.0 log CFU/coupon and they were statistically significant ($p < 0.05$). Moreover, it was observed that in both experimental systems the number of *E. coli* cells was higher than of L. *monocytogenes*. Initially, these differences were small, but from 17 h of biofilm formation, the number of cells of both strains differed significantly ($p < 0.05$).

3.2. Resistance to BAC of dual-species biofilm formed under batch and fed-batch conditions

The efficacy of BAC was evaluated in terms of reduction of viable adhered cells (RVAC). Reduction of viable adhered cells of L. *monocytogenes* and *E. coli* in dual-species biofilms formed under *batch* and *fed-batch* conditions with treatment of BAC-based disinfectant are showed in Fig. 3. The reduction levels were 2.18 and 2.20 log units per coupon and 1.72 and 0.69 log units per coupon for L. *monocytogenes* and *E. coli* under *batch* and *fed-batch* conditions, respectively. Both L. *monocytogenes* and *E. coli* achieved similar reduction levels when biofilm was formed in a *batch* system, while *E. coli* was definitely more resistant than L. *monocytogenes* in a *fed-batch* system ($p < 0.05$).

3.3. Microscopic images evaluation

The efficacy of BAC against dual-species biofilm of L. *monocytogenes-E. coli* was also evaluated using microscopic assays.

3.3.1. Epifluorescence microscopy – EFM

The ViaGram™ protocol allows for cell differentiation based on the integrity of the membranes in bacterial cells and based on Gram-signs. Live cells (interact membranes) stain blue (DAPI), while dead cells (damaged membranes) stain green (SYTOX) and Gram-positive cells stain red due to selective binding. The EFM images showed a denser biofilm structure and architecture under *batch* conditions when comparing with the *fed-batch* system (Fig. 4A and C). Under *batch* system, cells were organized into clusters, forming interconnected colonies, a significant part of the surface was covered with biofilm (Fig. 4A), while under *fed-batch* conditions, the cells were more dispersed, it was possible to observe free space as well as single cells whereas colonies and clusters are not dominant in the microscopic image (Fig. 4C). After BAC treatment, a clear damage in the structure of the biofilm formed under *batch* conditions can be appreciated in the images (Fig. 4B). Indeed, cell aggregates and single cells are more separated and so a less denser structure can be appreciated. Contrary, the structure of biofilm after BAC in *fed-batch* system changed relative to the control, but these changes were not as noticeable as in *batch* system. The main difference between two experimental set ups is the number of cells, not their location, which indicates the stages of biofilm development. In the case of the control and after BAC treatment in *fed-batch* system, we observe single cells, without aggregates and microcolonies. Looking at the proportion of cells (separately in control and after BAC treatment), after BAC, the blue-stained cells are superior to red cells (Fig. 4D).

3.3.2. Confocal laser scanning microscopy – CLSM

EFM study was complemented by the visualization of L. *monocytogenes-E. coli* stained with LIVE/DEAD under CLSM (Fig. 5). A general look at microscopic images of biofilms formed in a *batch* and *fed-batch* experimental system obtained with the use of LIVE/DEAD staining allowed to observe a dense, complex structure covering almost the entire field of view during the development of biofilm in a *batch* system, while in a *fed-batch* system a greater dispersion of cells was observed (Fig. 5A and C). Cells in the control biofilms showed differences depending on the conditions of biofilm development – in the *batch* system the cells had intact cytoplasmic membranes as evidenced by the green color of the cells (Fig. 5A), while in the *fed-batch* system the cells were characterized by damaged cytoplasmic membranes and absorption of both dyes, which made them greenish-yellowish color (Fig. 5C). This observation testifies the viability of bacterial cells. BAC inactivation affected the structure of the biofilm formed under *batch* conditions (Fig. 5B). Additionally, a subpopulation of dead cells (red), with damaged cytoplasmic membranes was distinguished. Moreover, the remaining cells were stained yellow, thus, despite the physiological activity of the cells, their cytoplasmic membranes were disturbed. Inactivation of the biofilm formed

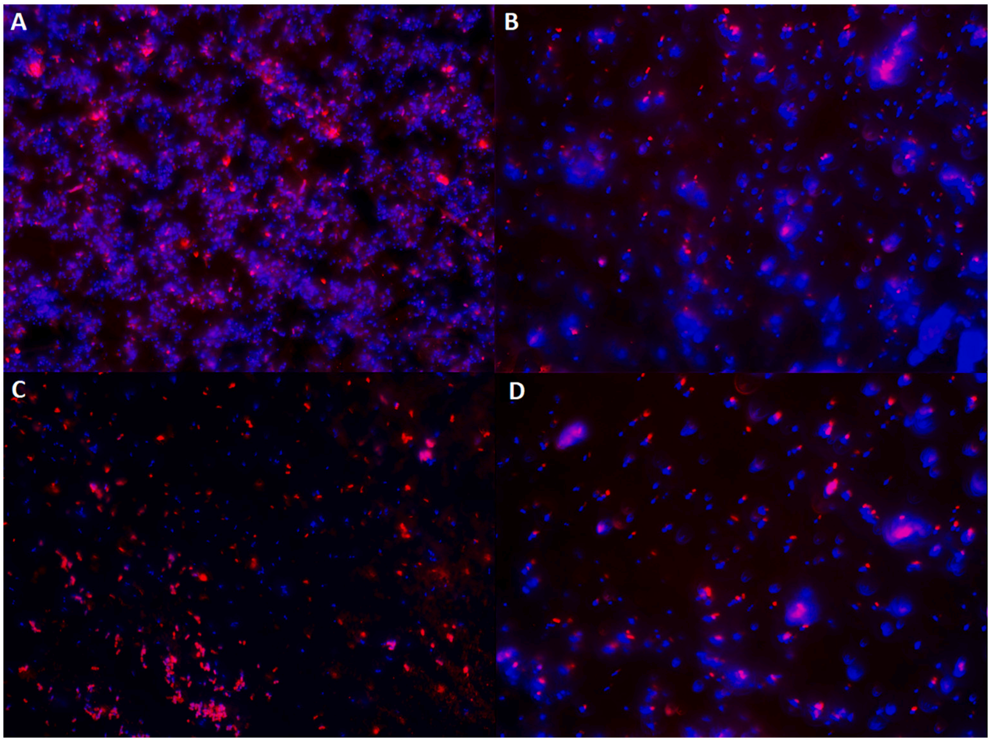

Fig. 4. Representative EFM images stained with ViaGram™ of *L. monocytogenes* A1 and *E. coli* A14 in 48 h old dual-species biofilms formed in *batch* (A, B) and *fed-batch* (C, D) system. Images represent cells in control biofilm (A, C) and after 10 min of BAC-based disinfectant treatment (B, D). Red cells represent live cells of L. *monocytogenes* A1 and blue cells represent live cells of *E. coli* A14 as described in Materials and methods according the procedure Via-Gram™. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

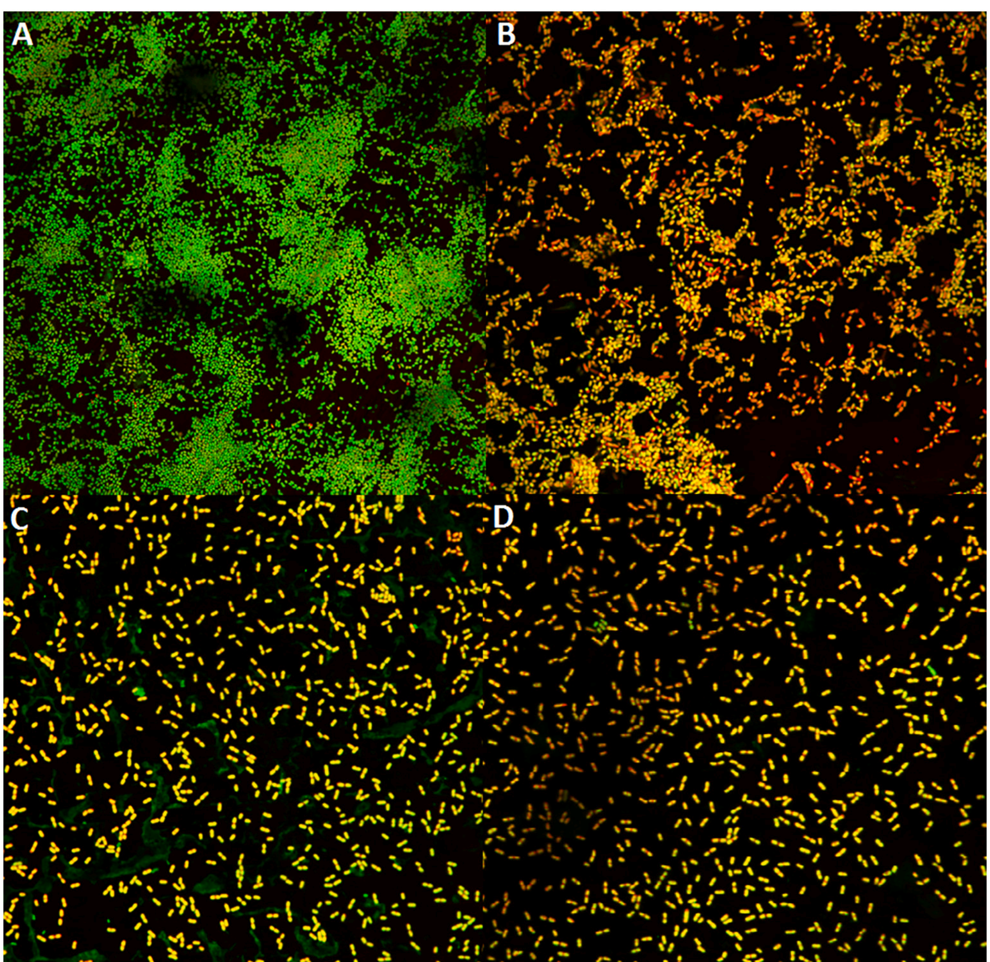

Fig. 5. Representative CLSM images stained with LIVE/DEAD viability kit of *L. monocytogenes* A1 and *E. coli* A14 in 48 h old dual-species biofilms formed in *batch* (A, B) and *fed-batch* (C, D) system. Images represent cells in control biofilm (A, C) and after 10 min of BAC-based disinfectant treatment (B, D). Images represent live (green) and dead (red) cells as described in Materials and methods. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

in the *fed-batch* system did not show the effect of BAC on the structure of the biofilm compared to the control (Fig. 5D). In turn, a slight decrease in the proportion of living cells (green), with intact cytoplasmic membranes, was observed. Remaining adhered cells were *greenish-yellow*ish, without a distinct subpopulation of cells with damaged membranes (red). This may indicate a higher resistance to BAC-based disinfectant of cells formed in a *fed-batch* system compared to a biofilm formed in a *batch* system.

4. Discussion

In the present study, the operational culture conditions in the context of BAC-resistance of L. *monocytogenes-E. coli* dual-species biofilm, was determined. The comparison of growth dynamic in two operational cultures demonstrated a higher number of cells in the biofilm formed in the *batch system* when comparing with the *fed-batch* system. In addition, it was shown that the number of *E. coli* was higher than L. *monocytogenes* regardless of the experimental set-up. These results are in agreement with previous studies carried out in our laboratory, in which higher *E. coli* adherence viable cell counts in mixed biofilms formed in stainless steel compared to L. *monocytogenes* were observed (Rodriguez-Lopez et al., 2017). Indeed, microscopic analyzes (EFM as well as CLSM) showed a higher cell density and a more developed biofilm structure under *batch* conditions.

Similar results regarding the adhesion of bacterial cells under static and dynamic conditions were obtained by Song et al. (2017), who cultured biofilm of periodontal pathogens on hydroxyapatite discs in test plates and in a CDC bioreactor. The amount of adhered bacterial cells, determined by scanning electron microscopy (SEM) and confocal laser scanning microscopy (CLSM) was higher in biofilms formed in a static system. The authors stated that this effect is because under static conditions, cell adhesion was due to the force of gravity, while in a dynamic system, cell adhesion is somehow hindered by shaking and for the vertical placing of the coupons. Interestingly, the authors showed no significant differences in thickness or resistance against chlorhexidine when comparing biofilms formed in a static and dynamic system. The limiting effect of hydrodynamic conditions on biofilm formation was shown by Fonseca and Sousa (2007). In their study, it was shown that the shear force (shaking) reduced the adhesion and biofilm formation capacity of *P. aeruginosa* compared to static conditions. In turn, Cotter et al. (2010) demonstrated a limiting effect of rotational speed on biofilm formation of *S. epidermidis*.

The above-mentioned studies indicate that the dynamic system achieved by mixing or shaking is a stressful condition for cells. This has consequences in hampering cell adhesion and slow biofilm formation, thus giving rise to a simpler 2D and 3D biofilm structure. But the most important results is that mixed biofilm formed under *fed-batch* conditions was more resistant to BAC, even although the number of cells in this biofilm was smaller and it had a less dense structure than biofilm formed in a *batch* system. Additionally, it was also observed that L. *monocytogenes* was more sensitive to BAC than *E. coli* under dynamic conditions, which is in agreement with previous studies carried out to characterize the effects of combined enzyme-BAC treatments against the removal of this dual-species biofilm (Rodriguez-Lopez and López-Cabo, 2017).

In general, in the literature it is accepted that complex-structured biofilms are more resistant (de Oliveira et al., 2010; Liu et al., 2017; Saá Ibusquiza et al., 2012; Sengupta et al., 2013). Several factors contribute to this general assumption: i) Biofilms acquire a complex 3D structure, in which cells of the outer layers protect the cells located in the inner layers to external stimuli and to the penetration of antimicrobial agents (Gebreyohannes et al., 2019), ii) Development of a intercellular communication system between bacteria inside the biofilm, quorum sensing. This cell-to-cell communication regulates the behavior of bacteria, including the regulation of gene expression responsible of cells resistance (Singh et al., 2017; Tang and Zhang, 2014), iii) Production of EPS (extracellular polymeric substances), which surrounds the biofilm and can also play a protective role (Uhlich et al., 2006) and iv) the effect of the presence of bacterial dead cells in the outer layers of the biofilm, which are most exposed to stress factors and protect the cells located in the inside of the biofilm structure (de Oliveira et al., 2010).

Contrary to this assumption, our results demonstrated no correlation between the complexity of the biofilm structure and its level of resistance to BAC. We hypothesize that *fed-batch* bioreactor operation implies limiting adhesion factors derived of the vertical orientation of the SS-coupons in presence of continuous gentle mixing that could promote an increase in individual cell resistance. In other words, at the time cells overcome unfavorable adhesion conditions they acquire individual resistance that could imply cross-resistance to BAC. This is in agreement with previous studies in which higher resistant biofilms were formed under dynamic conditions (Cotter et al., 2010; Fonseca and Sousa, 2007; Song et al., 2017). Moreover, fresh nutrient supply could have an impact on the physiological state of a single cell, shaping its *well-being* ("health"), which may contribute to the higher cell resistance to BAC in biofilms formed under *fed-batch* conditions. Furthermore, the supply of fresh medium could result not only in better "health" of individual bacterial cells in the biofilm, but also on the metabolic activity of the entire biofilm community, including parameters determining better resistance. The high metabolic activity of the biofilm formed under *fed-batch* conditions was stimulated by the inflow of fresh medium, which in turn increases the rate of extracellular polymeric substances (EPS) synthesis, regulation of the individual genes expression, as well as the synthesis of molecules participating in *quorum-sensing* (QS) (Flemming et al., 2007; Nadell et al., 2015; Wingender et al., 2001). By the contrary, the depletion of nutrients in biofilms formed in *batch* system might have a limiting effect on the metabolic activity of bacterial cells, and so they stay sensitive to BAC despite the complex 3D structure. The next factor which could determine the higher resistance of bacterial cells under *fed-batch* conditions is vertical alignment of coupons in CDC bioreactor. In order to produce biofilm on such a surface, bacterial cells have to overcome more factors than cells in *batch* system, where the force of gravity facilitates the adhesion of bacterial cells to the surface of the coupons. So, our hypothesis is that this "adhesion effort" of bacteria results in their higher resistance.

Our results unplug somehow the assumed relation between biofilm structure and biofilm resistance to external stimuli. Moreover, they lead to reflect on the meaning and implications of a classified "poor biofilm former bacteria" and highlight the importance of considering resistance related parameters for a correct evaluation of the risk associated to cross-contamination from biofilms formed in food industry surface.

5. Conclusions

In summary, our results indicated that dynamic conditions, represented in this study by the *fed-batch* system, limited the adhesion of bacterial cells, which affects the architecture and BAC-resistance of the L. *monocytogenes-E. coli* biofilm. Microscopic analysis of both EFM and CLSM showed a denser structure, composed of interconnected clusters in biofilms formed in *batch* system, while scattered and even single cells were observed in *fed-batch* system. However, biofilms formed in the *fed-batch* system were more resistant to BAC than biofilms formed in *batch* system. Our study indicates that a complex 3D structure does not always mean the higher resistance of biofilms. This observation sheds new light on the problem of biofilm in the food industry and indicates the need of considering operational conditions of the culture like hydrodynamic factors, shaking or the inflow of fresh medium and even concentration of the culture medium by e.g. mixing, which will better reflect the situation in food industry plants.

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CRediT authorship contribution statement

Aleksandra Maria Kocot (AMK) and Marta Lopez Cabo (MLC): conceptualization and methodology; AMK: carried out the experiments, performed data collection and analysis, interpreted and visualization the data, writing the original draft; MLC: scientific direction and coordination of the study, data analysis and discussion; MLC, Barbara Wróblewska (BW): reviewing and editing. All authors approved the final version of the manuscript.

Declaration of competing interest

The authors declare no conflict of interest.

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